How to Run the C++ Internship Assignment

WSL Ubuntu setup, run steps, and submission checklist

What this assignment does:

Part 1 uses GStreamer to read an MP4 video, resize frames to 640x640, and save them as JPEG images.

Part 2 uses a C++ OpenCV program to read those JPEG frames, detect faces with an OpenCV XML classifier, crop them, and save them in frame-wise folders.

1. Files needed

- input.mp4 – Download the input.mp4 and add to the folder : LINK
- run_part1.sh
- src/main.cpp
- Makefile
- haarcascade_frontalface_alt2.xml or haarcascade_frontalface_default.xml

2. Install tools in WSL Ubuntu

Run these commands in Ubuntu terminal:

- `sudo apt update`
- `sudo apt install -y g++ make pkg-config unzip`
- `sudo apt install -y libopencv-dev`
- `sudo apt install -y gstreamer1.0-tools gstreamer1.0-plugins-base gstreamer1.0-plugins-good gstreamer1.0-plugins-bad gstreamer1.0-libav`

3. Create project folder

- `mkdir -p ~/cpp_internship_assignment/src`
- `cd ~/cpp_internship_assignment`

Copy your files into this folder. Put the video file as input.mp4.

4. Run Part 1

Make the script executable and run it:

- `chmod +x run_part1.sh`
- `./run_part1.sh`

Expected result: a frames folder is created with files like frame_00000.jpg, frame_00001.jpg, and so on.

5. Compile and run Part 2

- `make`
- `./face_cropper`

If alt2 XML is not available, run the program with the default XML classifier:

- `./face_cropper frames cropped_faces haarcascade_frontalface_default.xml`

Expected result: a cropped_faces folder is created. Inside it, each frame gets its own folder, and every detected face is saved as face_1.jpg, face_2.jpg, and so on.

6. Final folder structure

```
cpp_internship_assignment/
├── input.mp4
├── run_part1.sh
├── Makefile
├── src/
│   └── main.cpp
├── frames/ ..
└── cropped_faces/..
```